

POLU NANDEESWAR REDDY

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Professional Summary

AWS DevOps Engineer with 2+ years owning end-to-end infrastructure for 5 live production SaaS platforms on AWS ECS/EC2, supporting 20+ engineers across 3 teams. Achieved 99.95% uptime, MTTR under 20 minutes, and ~60% annual SaaS cost savings by self-hosting GitLab, SonarQube, OpenProject, and FlexHR on self-managed AWS infrastructure. Reduced deployment time by ~80% via end-to-end AWS CodePipeline automation and cut idle compute spend by ~40%/month through multi-account Lambda cost controls and FinOps governance. Currently advancing into AWS EKS and Kubernetes orchestration.

Technical Skills

AWS: VPC, EC2, IAM, RDS, S3, IAM, Lambda, Route 53, ECS, ECR, Auto Scaling Groups, CloudWatch, AWS Organizations, Identity Center, EFS, EventBridge, Guardrails, CloudFormation, Amazon Sagemaker, Bedrock, EKS, AWS CDK, Systems Manager

Containers & Orchestration: Docker, AWS ECS/ECR, Docker Compose, Kubernetes & EKS (In Progress)

Infrastructure as Code: Terraform, CloudFormation, AWS CDK

CI/CD: AWS CodePipeline, GitLab CI/CD, GitHub Actions, Jenkins,

Security: IAM, Security Groups, VPC Design, AWS Organizations, AWS WAF, GuardDuty, RBAC

Monitoring & Observability: CloudWatch, Prometheus, Grafana

Scripting & Automation: Python, Bash, Lambda Functions, Event-Driven Architecture

Version Control & Code Quality: Git, GitLab, SonarQube, GitHub

Databases & Caching: RDS, PostgreSQL, Redis, MySQL

Application Development: React.js, Spring Boot

Professional Experience

AWS DevOps Engineer

Yotta Tech Ports Inc

Dec 2024 – Present

Hyderabad, India

- **Generative AI & Chatbot Integration:** Architected and deployed a production-grade AI-powered customer support chatbot on the company website leveraging **Amazon Bedrock** (Claude foundation model) — integrating API Gateway for a fully serverless inference pipeline, reducing customer support response time by an estimated ~70% and deflecting repetitive tier-1 queries from the support team
- **Cloud Infrastructure & Multi-Application Hosting:** Sole DevOps owner for 5 live production applications (GitLab, SonarQube, OpenProject, FlexHR, and a custom internal tool) hosted on AWS ECS and EC2 — supporting 3 development teams across 20+ engineers with zero dedicated infra team
- Provisioned and maintained full AWS infrastructure per application — VPC, subnets, EFS, RDS, ALB, security groups — cutting external vendor dependency and estimated SaaS licensing costs by ~60% annually by running self-managed open-source alternatives
- All 4 platforms are live and accessible in production: git.yottatechports.com (GitLab), sonar.yottatechports.com (SonarQube), agile.yottatechports.com (OpenProject), flexhr.yottatechports.com (FlexHR)
- Managed 5 custom subdomain configurations on Route 53 (CNAME, A records) with zero unplanned DNS downtime across all changes
- **Security & Governance:** Built a Lambda-based auto-remediation function that detects and removes overly permissive security groups (0.0.0.0/0) — reducing manual security audits by ~90% and cutting exposure window from days to near-zero
- **DevSecOps CI/CD Pipeline:** Embedded DevSecOps practices into CI/CD pipelines by integrating automated SAST/DAST scanning (SonarQube), secret detection, and image vulnerability scanning into the build stage — shifting security left and reducing vulnerabilities reaching production by an estimated ~70%
- **Hardened Container Images:** Hardened container images across 5 applications by authoring multi-stage Dockerfiles with minimal base images, non-root users, and pinned dependencies — cutting image attack surface and CVE count by ~50% while keeping images lean
- **Secure 3-Tier VPC Design:** Designed a secure 3-tier VPC architecture (public/private/data subnets) with least-privilege security groups, NACLs, and isolated RDS in private subnets — enforcing network segmentation and zero direct internet exposure for application and database tiers
- Managed AWS IAM Identity Center for 3+ project teams — created roles, groups, and least-privilege policies for every new project onboard, ensuring zero over-permissioned access in production accounts
- Secured all application secrets in AWS Secrets Manager across 5 applications, eliminating hardcoded credentials entirely from codebases
- **Incident Response & Reliability:** Maintained MTTR under 20 minutes across all production incidents, with an average initial response time of under 5 minutes from alert to active triage — across HTTP 500/502 failures on live applications used daily by 20+ engineers

- Authored RCA playbooks for every major incident, reducing repeat resolution time by an estimated ~70% for similar failure patterns across the 5 hosted apps
- **Cost Optimization & FinOps:** Built a multi-account Lambda automation that enforces `stop=true` tagging to shut down idle EC2 instances daily — estimated to reduce idle compute spend by ~40% per month across accounts, with automated SNS notifications to stakeholders
- Conducted monthly AWS billing reviews producing per-service cost reports; flagged and resolved abnormal spend spikes across 3+ AWS accounts, preventing unnoticed cost overruns
- Automated EC2 and RDS shutdown for untagged resources via Lambda — enforcing compliance and eliminating costs from orphaned infrastructure
- **Infrastructure as Code & Automation:** Used Terraform to manage Lambda runtime upgrades across multiple functions — reducing manual update effort by ~75% per release cycle and maintaining IaC consistency
- Created IAM users and permission boundaries for multiple new AWS onboardings, reducing access provisioning turnaround from days to under 30 minutes per engineer
- **Kubernetes & EKS (Shadowing):** Shadowing a senior-led AWS EKS migration by studying production cluster setup, RBAC, network policies, and resource limits — building hands-on Kubernetes orchestration skills to support secure, namespace-isolated workloads across staging and production

Junior DevOps Engineer

Jun 2024 – Dec 2024

Yotta Tech Ports Inc

Hyderabad, India

- **GitLab Self-Hosted on AWS ECS — Built From Zero:** Designed and delivered a production-grade self-managed GitLab on AWS ECS using Docker — from a proof of concept to a live environment serving 20+ developers across 3 teams, eliminating the need for a paid GitLab SaaS subscription and saving an estimated ~\$3,000–\$5,000/year in licensing costs
- Engineered a full AWS CodePipeline (source trigger → manual approval → build → ECS deploy) that reduced deployment effort from manual steps to a single approval click, cutting deployment time by ~80%
- Built a custom Dockerfile and `buildspec.yml` automating image build, ECR push, and artifact storage in JSON — reducing build configuration effort by ~65% compared to manual image management
- Wrote a `fetchsecrets.sh` script that dynamically injects AWS Secrets Manager credentials into GitLab's Rails config at container startup — removing 100% of hardcoded secrets from the deployment
- Provisioned complete AWS infrastructure from scratch for GitLab: ECS cluster, task definition, ECS service, VPC, 2 subnets, internet gateway, security groups, EFS, RDS, and ALB — delivered end-to-end in under 4 weeks as a first-time solo infrastructure build
- Configured weekly AWS Backup on EFS protecting all GitLab repository data — establishing a recovery point objective (RPO) of 7 days with zero manual backup effort
- Built a CloudWatch + EventBridge metrics dashboard with alerts across 10+ infrastructure metrics, providing real-time visibility into GitLab container and database health

Education

Bachelor of Technology – Electrical and Electronics Engineering

2020 – May 2023

Rajeev Gandhi Memorial College of Engineering and Technology

Nandyal, Andhra Pradesh

- 75% – Graduated with Distinction

Certifications & Professional Development

In Progress: AWS Certified Solutions Architect – Associate, AWS Certified DevOps Engineer – Professional

Active Training: Kubernetes Administration (Udemy), AWS EKS Deep Dive, Container Orchestration

Completed: Java Full Stack Developer, Terraform Associate, Docker Mastery, Python for DevOps